CHE110: Environmental Studies

Key Facts, Laws, Standards & Quick Reference Handbook

Faculty of Environmental Sciences & Engineering

Edition 2026–2027

Contents

CHAPTER 1

Unit 1: Introduction to Environment, Sustainability Metrics & SDGs

Atmospheric, Lithospheric & Hydrospheric Invariants

- **Four Environmental Spheres**:
  - **Atmosphere**: Total mass $\approx 5.15 \times 10^{18}$ kg. Dry air composition by volume: $N_2 = 78.08\%$, $O_2 = 20.95\%$, $Ar = 0.93\%$, $CO_2 = 0.042\%$ (420 ppm), trace gases.
  - **Atmospheric Layers**:
    - **Troposphere** (0–12 km): Contains 80% atmospheric mass; temperature decreases at Environmental Lapse Rate (ELR $\approx 6.5°$C/km).
    - **Stratosphere** (12–50 km): Temperature inversion due to ozone UV absorption; Peak ozone layer at 20–30 km.
    - **Mesosphere** (50–85 km): Coldest layer (down to $-90°$C); meteor ablation.
    - **Thermosphere** (85–600 km): High kinetic temperatures from solar X-ray/EUV photoionization.
  - **Hydrosphere**: Global water volume $\approx 1.386 \times 10^9$ km^3. Distribution: Saline ocean water = 97.5%, Freshwater = 2.5%.
  - **Freshwater Partitioning**: Glaciers and polar ice caps = 68.7%, Groundwater = 30.1%, Surface water (lakes, rivers, swamps) = 1.2%.
  - **Lithosphere**: Continental crust (granitic, 30–50 km thick, density ≈ 2.7 g/cm^3) vs Oceanic crust (basaltic, 5–10 km thick, density ≈ 3.0 g/cm^3).

Environmental Impact & Carrying Capacity Formulas

- **Ehrlich-Holdren IPAT Identity**:
$$I = P \times A \times T \tag{1.1}$$
where I = Environmental Impact, P = Population size, A = Affluence (GDP or consumption per capita), T = Technology (impact per unit consumption).
- **ImPACT Identity**:
$$I = P \times A \times C \times T' \tag{1.2}$$
where C = energy/resource consumption per unit GDP, and T' = pollution per unit energy consumed.
- **Ecological Footprint (EF) vs Biocapacity (BC)**:
$$\text{Ecological Deficit/Reserve} = BC - EF \tag{1.3}$$
If $EF > BC$, humanity is in global ecological overshoot (1.75 Earth equivalents currently).
- **Brundtland Definition of Sustainable Development (1987)**: *"Development that meets the needs of the present without compromising the ability of future generations to meet their own needs."* (Report: *Our Common Future*).
- **Three Pillars of Sustainability (Triple Bottom Line)**: Environmental (Planet), Social (People), and Economic (Profit).
- **Weak vs Strong Sustainability**:
  - **Weak Sustainability**: Assumes natural capital (K_n) and manufactured human capital (K_m) are fully fungible substitutes ($K_{\text{total}} = K_n + K_m = \text{constant}$).

– **Strong Sustainability**: Demands non-substitutability; critical natural capital (K_n^*) cannot be replaced by manufactured physical or human capital.

The 17 UN Sustainable Development Goals (SDGs 2016–2030)

- **SDG 1**: No Poverty • **SDG 2**: Zero Hunger • **SDG 3**: Good Health and Well-being
- **SDG 4**: Quality Education • **SDG 5**: Gender Equality
- **SDG 6**: **Clean Water and Sanitation** (Safe drinking water, wastewater recycling, IWRM).
- **SDG 7**: **Affordable and Clean Energy** (Doubling renewable energy share, energy efficiency).
- **SDG 8**: Decent Work and Economic Growth • **SDG 9**: Industry, Innovation and Infrastructure
- **SDG 10**: Reduced Inequalities
- **SDG 11**: **Sustainable Cities and Communities** (Air quality, transit, municipal waste).
- **SDG 12**: **Responsible Consumption and Production** (Circular economy, 3R, food loss).
- **SDG 13**: **Climate Action** (Paris Agreement Nationally Determined Contributions - NDCs).
- **SDG 14**: **Life Below Water** (Marine conservation, halting ocean acidification and plastic pollution).
- **SDG 15**: **Life on Land** (Halt deforestation, combat desertification, halt biodiversity loss).
- **SDG 16**: Peace, Justice and Strong Institutions • **SDG 17**: Partnerships for the Goals.

Examiner's High-Yield Memory Tip & Quick Formula Shortcut

Remember the key environmental SDGs with the mnemonic **"W-E-C-C-C-L-L"**: Water (6), Energy (7), Cities (11), Consumption (12), Climate (13), Life below water (14), Life on land (15)!

CHAPTER 2

Unit 2: Natural Resources, Ecosystem Energetics & Succession

Resource Classifications & Depletion Parameters

- **Renewability Criterion**: A resource is renewable if replenishment rate $R_{\text{regen}} \geq$ Harvest rate H_{draw}. If $H_{\text{draw}} > R_{\text{regen}}$, resource exhaustion occurs.
- **Soil Formation Rate**: Pedogenesis generates approximately 1 cm of fertile topsoil every 200–400 years.
- **Universal Soil Loss Equation (USLE)**:

$$A = R \cdot K \cdot LS \cdot C \cdot P \tag{2.1}$$

  where A = computed soil loss, R = rainfall erosivity factor, K = soil erodibility factor, LS = slope length-gradient factor, C = crop cover factor, P = conservation practice factor.

- **Coastal Aquifer Saltwater Intrusion (Ghyben-Herzberg Relation)**:

$$z = \frac{\rho_f}{\rho_s - \rho_f} h \approx 40h \tag{2.2}$$

  where h = freshwater head above sea level, z = depth to saltwater interface below sea level, ρ_f = 1.000 g/cm^3, ρ_s = 1.025 g/cm^3. Every 1 meter drop in groundwater table induces a 40 meter saltwater rise.

- **National Forest Policy Targets (1988)**:
  - National target: Minimum 33% of total geographical land area under forest and tree cover.
  - Hill and mountainous ecosystems: Minimum 67% (2/3rd) forest cover for slope stability and soil conservation.
- **Wind Power Available Kinetic Energy**:

$$P_{\text{wind}} = \frac{1}{2}\rho A v^3 \tag{2.3}$$

  where ρ = air density (≈ 1.225 kg/m^3), A = swept rotor area (πR^2), v = wind velocity. Power scales with the cube of wind velocity (v^3). Betz limit maximum aerodynamic efficiency = 59.3% (16/27).

Ecosystem Energetics & Trophic Transfer Efficiencies

- **Primary Productivity Formulations**:

$$\text{GPP} = \text{Total rate of organic chemical energy synthesized by autotrophs via photosynthesis} \tag{2.4}$$
$$\text{NPP} = \text{GPP} - R_a \tag{2.5}$$

  where R_a = autotrophic respiration energy expended for cellular maintenance.

- **Net Ecosystem Productivity (NEP)**:

$$\text{NEP} = \text{GPP} - (R_a + R_h) = \text{NPP} - R_h \tag{2.6}$$

  where R_h = heterotrophic respiration (consumers + decomposers).

- **Lindeman's 10% Trophic Efficiency Rule (1942)**:

$$\text{Ecological Efficiency } (\lambda) = \frac{E_n}{E_{n-1}} \times 100\% \approx 10\% \tag{2.7}$$

  Approximately 90% of energy is lost at each trophic link via metabolic respiration (R), unconsumed biomass, and heat dissipation according to the Second Law of Thermodynamics.

- **Ecological Pyramids Comparison**:
  - **Pyramid of Numbers**: Upright in grasslands; inverted in parasitic food chains; spindle-shaped in temperate forest trees.
  - **Pyramid of Biomass**: Upright in terrestrial forests; **inverted in marine open waters** (phytoplankton standing crop is smaller than zooplankton due to rapid turnover).
  - **Pyramid of Energy**: **ALWAYS UPRIGHT** in all ecosystems without exception.

Ecological Succession Dynamics

- **Primary vs Secondary Succession**:
  - **Primary Succession**: Initiates on sterile, virgin substrates lacking soil (bare rock, cooled volcanic basalt, sand dunes, retreating glaciers). Pioneers: Lichens, mosses.
  - **Secondary Succession**: Occurs on disturbed sites where pre-existing soil and seed banks remain (abandoned farmland, burned forest). Significantly faster rate.
- **Six Seral Stages of Succession (Clementsian Model)**:
  1. **Nudation**: Development of bare exposed substrate.
  2. **Invasion / Migration**: Dispersal and arrival of propagules (seeds, spores).
  3. **Ecesis**: Successful establishment, germination, and growth of pioneers.
  4. **Aggregation**: Population increase and clustering of established species.
  5. **Competition**: Interspecific and intraspecific struggle for space, nutrients, and light.
  6. **Reaction & Climax Stabilization**: Biotic modification of soil and microclimate, culminating in a stable self-perpetuating climax community.
- **Trends from Pioneer to Climax**:
  - Gross production / community respiration (P/R ratio): $P/R > 1$ in early seral stages; $P/R \to 1.0$ at climax equilibrium.
  - Biomass: Low $\to$ High. Diversity (H'): Low $\to$ High. Food webs: Linear chains $\to$ Complex webs.

Examiner's High-Yield Memory Tip & Quick Formula Shortcut

The pyramid of energy is NEVER inverted! In open oceans, the biomass pyramid is inverted because phytoplankton reproduce and die in days, but the energy throughput per year is always higher at the base!

CHAPTER 3

Unit 3: Biodiversity Indices, Conservation & Protected Areas

Quantitative Diversity Indices

- **Species Richness (S)**: Total absolute number of distinct species recorded in a defined sample or biotope.
- **Shannon-Wiener Information Diversity Index (H')**:

$$H' = -\sum_{i=1}^{S} p_i \ln(p_i) \tag{3.1}$$

  where $p_i = n_i/N$ is the relative abundance proportion of species i.

- **Maximum Diversity & Pielou's Evenness (J')**:

$$H'_{\max} = \ln(S) \tag{3.2}$$

$$J' = \frac{H'}{H'_{\max}} = \frac{H'}{\ln(S)}, \quad 0 \leq J' \leq 1.0 \tag{3.3}$$

- **Simpson's Dominance Index (D)**:

$$D = \sum_{i=1}^{S} p_i^2 = \frac{\sum_{i=1}^{S} n_i(n_i - 1)}{N(N-1)} \tag{3.4}$$

- **Simpson's Index of Diversity ($1 - D$)** and **Reciprocal Simpson ($1/D$)**.
- **Alpha, Beta, and Gamma Diversity (Whittaker, 1960)**:

$$\alpha\text{-diversity} = \text{Local within-habitat species richness} \tag{3.5}$$

$$\beta\text{-diversity} = \frac{\gamma}{\bar{\alpha}} = \text{Species turnover rate between habitats across gradients} \tag{3.6}$$

$$\gamma\text{-diversity} = \text{Total landscape or regional biodiversity} \tag{3.7}$$

Hotspots, Megadiversity & Biogeographic Realms

- **Myers Biodiversity Hotspot Criteria (2000)**:
  1. Must contain at least 1,500 endemic species of vascular plants ($> 0.5\%$ of world total).
  2. Must have lost at least 70% of its primary original native vegetation ($\leq 30\%$ remaining intact).
- **The 4 Indian Biodiversity Hotspots**:
  1. **Western Ghats and Sri Lanka**: Exceptionally high amphibian, reptilian, and floral endemism.
  2. **Himalaya**: Entire Indian Himalayan mountain arc (spanning Jammu & Kashmir, HP, Uttarakhand, Sikkim, Arunachal Pradesh).
  3. **Indo-Burma**: Covers North-Eastern states south of the Brahmaputra River, Andaman Islands.
  4. **Sundaland**: Includes the Nicobar group of islands (extending through Malaysia and Indonesia).
- **Megadiverse Countries**: India is recognized among the 17 megadiverse nations, harboring $\approx$ 7–8% of global recorded floral and faunal species across only 2.4% of global terrestrial land area.

- **Biogeographic Zones of India (Rodgers & Panwar Classification - 10 Zones)**: 1. Trans-Himalaya, 2. Himalaya, 3. Desert, 4. Semi-Arid, 5. Western Ghats, 6. Deccan Peninsula, 7. Gangetic Plain, 8. Coasts, 9. North-East, 10. Islands.

IUCN Categories, Protected Area Frameworks & Statutes

- **IUCN Red List Categories**: Extinct (EX), Extinct in the Wild (EW), **Critically Endangered (CR)**, **Endangered (EN)**, **Vulnerable (VU)**, Near Threatened (NT), Least Concern (LC), Data Deficient (DD).
- **Quantitative Thresholds for Threatened Categories**:

IUCN Quantitative Criterion	Critically Endangered (CR)	Endangered (EN)	Vulnerable (VU)
Population reduction over 10 yrs	$\geq 90\%$ (observed)	$\geq 70\%$	$\geq 50\%$
Extent of Occurrence (EOO)	< 100 km^2	$< 5{,}000$ km^2	$< 20{,}000$ km^2
Area of Occupancy (AOO)	< 10 km^2	< 500 km^2	$< 2{,}000$ km^2
Number of mature individuals	< 50	< 250	$< 1{,}000$

- **In-Situ vs Ex-Situ Conservation**:
  - **In-Situ**: National Parks (no human grazing/habitation), Wildlife Sanctuaries (limited rights permitted), Biosphere Reserves (Core, Buffer, Transition zones), Tiger Reserves, Sacred Groves.
  - **Ex-Situ**: Zoological gardens, Botanical gardens, Seed banks, Cryopreservation, In-vitro gene banks.
- **Key Wildlife Statutes**: Wildlife (Protection) Act, 1972 (amended 2022); Biological Diversity Act, 2002 (National Biodiversity Authority - NBA, State Biodiversity Boards - SBB, Biodiversity Management Committees - BMC).

CHAPTER 4

Unit 4: Environmental Pollution Standards, AQI, Water Quality & Waste

CPCB National Ambient Air Quality Standards (NAAQS 2009)

Pollutant	Time-Weighted Average	Industrial, Residential & Rural	Ecologically Sensitive
Particulate Matter (PM_{10})	Annual / 24 Hours	60 / 100 $\mu g/m^3$	60 / 100 $\mu g/m^3$
Fine Particulate ($PM_{2.5}$)	Annual / 24 Hours	40 / 60 $\mu g/m^3$	40 / 60 $\mu g/m^3$
Sulphur Dioxide (SO_2)	Annual / 24 Hours	50 / 80 $\mu g/m^3$	20 / 80 $\mu g/m^3$
Nitrogen Dioxide (NO_2)	Annual / 24 Hours	40 / 80 $\mu g/m^3$	30 / 80 $\mu g/m^3$
Carbon Monoxide (CO)	8 Hours / 1 Hour	2.0 / 4.0 mg/m^3	2.0 / 4.0 mg/m^3
Ozone (O_3)	8 Hours / 1 Hour	100 / 180 $\mu g/m^3$	100 / 180 $\mu g/m^3$
Lead (Pb)	Annual / 24 Hours	0.50 / 1.0 $\mu g/m^3$	0.50 / 1.0 $\mu g/m^3$
Ammonia (NH_3)	Annual / 24 Hours	100 / 400 $\mu g/m^3$	100 / 400 $\mu g/m^3$

National Air Quality Index (AQI) Categories:

- 0–50: Good 51–100: Satisfactory 101–200: Moderate 201–300: Poor 301–400: Very Poor 401–500: Severe.

Water Quality Parameters, BOD Kinetics & Streeter-Phelps

- **Biochemical Oxygen Demand (BOD Kinetics)**:

$$y(t) = L_0(1 - e^{-k_d t}) \tag{4.1}$$

$$\text{BOD}_5 = L_0(1 - e^{-5k_d}) \approx 0.68 L_0 \quad (\text{for standard } k_d \approx 0.23 \text{ day}^{-1}) \tag{4.2}$$

  where L_0 = ultimate carbonaceous BOD, $y(t)$ = BOD exerted at time t, k_d = deoxygenation constant.

- **Chemical Oxygen Demand (COD)**: Measures total oxygen required for chemical oxidation by acidic potassium dichromate ($K_2Cr_2O_7$). COD $\geq$ BOD_u $\geq$ BOD_5. $\text{BOD}_5/\text{COD} \geq 0.5$ implies high biodegradability.

- **Streeter-Phelps Oxygen Sag Equation**:

$$D(t) = \frac{k_d L_0}{k_r - k_d}\left(e^{-k_d t} - e^{-k_r t}\right) + D_0 e^{-k_r t} \tag{4.3}$$

where $D(t)$ = DO deficit at time t, D_0 = initial DO deficit, k_r = stream reaeration rate constant.

- **Critical Time (t_c) and Critical Deficit (D_c)**:

$$t_c = \frac{1}{k_r - k_d} \ln \left\{ \frac{k_r}{k_d} \left[1 - D_0 \frac{k_r - k_d}{k_d L_0} \right] \right\} \tag{4.4}$$

$$D_c = \frac{k_d}{k_r} L_0 e^{-k_d t_c} \tag{4.5}$$

- **CPCB Designated Best Use Classes**: Class A (Drinking source without treatment, DO $\geq$ 6 mg/L, BOD $\leq$ 2 mg/L); Class D (Aquatic fisheries propagation, DO $\geq$ 4.0 mg/L, Free ammonia $\leq$ 1.2 mg/L).

Noise Standards & Solid Waste Management Rules

- **Ambient Noise Standards (Noise Rules, 2000)**:

Zone Code	Category of Area	Day Limit (6 AM – –10 PM)	Night Limit (
(A)	Industrial Area	75 dB(A)	70
(B)	Commercial Area	65 dB(A)	55
(C)	Residential Area	55 dB(A)	45
(D)	Silence Zone (within 100 m of hospitals, schools)	50 dB(A)	40

- **Solid Waste Hierarchy**: Refuse → Reduce → Reuse → Repurpose → Recycle → Recovery → Landfill Disposal.
- **Biomedical Waste Color Coding (2016 Rules)**:
  - **Yellow**: Human anatomical, animal waste, soiled dressings (Incineration / Plasma pyrolysis).
  - **Red**: Contaminated recyclable plastics, catheters, tubing, IV bottles (Autoclaving → Shredding).
  - **White (Translucent)**: Sharps, needles, scalpels (Autoclave/Dry heat → Sharp pit).
  - **Blue**: Glassware, medicine vials, metallic body implants (Disinfection with sodium hypochlorite).

CHAPTER 5

Unit 5: Disaster Risk Reduction, Hazard Scales & Sendai Framework

Disaster Risk Formulas & Hazard Measurement Scales

- **Universal Disaster Risk Formulation (UNDRR)**:

$$\text{Disaster Risk} = \frac{\text{Hazard} \times \text{Exposure} \times \text{Vulnerability}}{\text{Coping Capacity}} \tag{5.1}$$

- **Earthquake Magnitude & Energy Release**:
  - **Richter Local Magnitude (M_L)**:

$$M_L = \log_{10}(A) - \log_{10}(A_0) \tag{5.2}$$

    A unit increase in magnitude represents a 10-fold increase in ground motion amplitude.
  - **Gutenberg-Richter Energy Release Equation**:

$$\log_{10} E = 4.8 + 1.5M \implies E = 10^{4.8} \times 10^{1.5M} \tag{5.3}$$

    Each unit increase in Richter magnitude corresponds to a $10^{1.5} \approx 31.62$-fold increase in radiated seismic strain energy.
  - **Modified Mercalli Intensity (MMI)**: Qualitative 12-point scale (I: Not felt to XII: Total destruction) evaluating ground shaking and structural damage.
- **IMD Cyclone Classification (Maximum Sustained Wind Speeds - 3 min avg)**:
  - Depression: 31–49 km/h (17–27 knots)
  - Deep Depression: 50–61 km/h (28–33 knots)
  - Cyclonic Storm: 62–88 km/h (34–47 knots)
  - Severe Cyclonic Storm: 89–117 km/h (48–63 knots)
  - Very Severe Cyclonic Storm: 118–166 km/h (64–89 knots)
  - Extremely Severe Cyclonic Storm: 167–221 km/h (90–119 knots)
  - Super Cyclonic Storm: ≥ 222 km/h (≥ 120 knots)

Disaster Management Act, 2005 & Sendai Framework (2015–2030)

- **Three-Tier Institutional Architecture (DM Act, 2005)**:
  - **National Level**: National Disaster Management Authority (NDMA) headed by the Prime Minister. National Executive Committee (NEC) headed by Home Secretary.
  - **State Level**: State Disaster Management Authority (SDMA) headed by the Chief Minister.
  - **District Level**: District Disaster Management Authority (DDMA) headed by the District Magistrate / Collector.
  - **Specialist Agencies**: National Disaster Response Force (NDRF) - dedicated multi-disciplinary battalions; National Institute of Disaster Management (NIDM) for training and capacity building.
- **The 4 Sendai Priorities for Action (2015–2030)**:
  1. Priority 1: Understanding disaster risk.

2. Priority 2: Strengthening disaster risk governance to manage disaster risk.
3. Priority 3: Investing in disaster risk reduction for resilience.
4. Priority 4: Enhancing disaster preparedness for effective response and to **"Build Back Better"** in recovery, rehabilitation and reconstruction.

- **The 7 Global Sendai Targets (A to G)**: Reduce mortality (A), reduce affected people (B), reduce direct economic loss (C), reduce infrastructure damage (D), increase national/local DRR strategies by 2020 (E), enhance international cooperation (F), increase access to multi-hazard early warning systems (G).

Examiner's High-Yield Memory Tip & Quick Formula Shortcut

Remember that each +1 increase on the Richter scale means ≈ 31.6 times more energy release, and a +2 increase means $31.62^2 = 1000$ times more energy release!

CHAPTER 6

Unit 6: Environmental Demography, Constitutional Law & Movements

Demographic Growth Models & Doubling Time

- **Exponential Population Growth Model (Malthusian)**:

$$\frac{dN}{dt} = rN \tag{6.1}$$

$$N(t) = N_0 e^{rt} \tag{6.2}$$

  where N_0 = initial population, r = intrinsic per-capita rate of increase $(b - d)$, t = time elapsed.

- **Rule of 70 (Doubling Time)**:

$$T_d = \frac{\ln(2)}{r} = \frac{0.69315}{r} \approx \frac{70}{\text{Growth Rate in Percentage } (r\%)} \tag{6.3}$$

  For example, if a population grows at $r = 2.0\%$ per annum, its doubling time is $T_d = 70/2 = 35$ years.

- **Logistic Population Growth Model (Verhulst-Pearl)**:

$$\frac{dN}{dt} = rN\left(1 - \frac{N}{K}\right) \tag{6.4}$$

  where K = environmental carrying capacity.

  - When $N \ll K$, $dN/dt \approx rN$ (exponential phase).
  - When $N = K/2$, growth rate dN/dt reaches its absolute maximum (inflection point).
  - When $N \to K$, $dN/dt \to 0$ (asymptotic plateau).

- **Integrated Logistic Equation**:

$$N(t) = \frac{K}{1 + \left(\frac{K - N_0}{N_0}\right) e^{-rt}} \tag{6.5}$$

- **Demographic Transition Model (DTM)**:

  - Stage 1 (Pre-Industrial): High birth rate, high death rate $\implies$ Slow growth.
  - Stage 2 (Transitional): High birth rate, plunging death rate $\implies$ Rapid population explosion.
  - Stage 3 (Industrial): Declining birth rate, low death rate $\implies$ Decelerating growth.
  - Stage 4 (Post-Industrial): Low birth rate, low death rate $\implies$ Population stabilization or contraction.

Chronology of Major Environmental Movements in India

- **Bishnoi Movement (1730)**: Khejarli, Rajasthan. Amrita Devi and 363 Bishnois sacrificed lives protecting Khejri (*Prosopis cineraria*) trees.
- **Chipko Movement (1973)**: Chamoli, Uttarakhand. Led by Gaura Devi, Sunderlal Bahuguna, Chandi Prasad Bhatt. Villagers hugged trees to prevent commercial clear-cutting. Resulted in 15-year Himalayan commercial logging ban (1980).
- **Silent Valley Movement (1978)**: Kerala. Led by Kerala Sastra Sahitya Parishad (KSSP) to save

evergreen rainforests from a hydroelectric dam on the Kunthipuzha River. Led to creation of Silent Valley National Park (1984).

- **Appiko Movement (1983)**: Uttara Kannada, Karnataka. Led by Pandurang Hegde; southern Indian counterpart of Chipko.
- **Jungle Bachao Andolan (1982)**: Singhbhum, Bihar (now Jharkhand). Adivasi resistance against replacing native Sal forests with commercial Teak monocultures.
- **Narmada Bachao Andolan (1985–present)**: Led by Medha Patkar and Baba Amte opposing mega-dams (Sardar Sarovar) on the Narmada River, demanding rehabilitation and resettlement rights.